

Assignment 4

Addition

	5	4	3	2	1	0	0	i
addendExpon				14	8	0		

$$3x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	0	i
addendCoef				3	-2	1		

addendSize

3

	5	4	3	2	1	0	0	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	0	j
adderCoef				8	-3	10		

adderSize

3

	5	4	3	2	1	0	0	k
sumExpon								

	5	4	3	2	1	0	0	k
sumCoef								

sumSize

	5	4	3	2	1	0	0	i
addendExpon				14	8	0		

$$3x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	0	i
addendCoef				3	-2	1		

addendSize

3

	5	4	3	2	1	0	0	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	0	j
adderCoef				8	-3	10		

adderSize

3

	5	4	3	2	1	0	0	k
sumExpon						0		

1

	5	4	3	2	1	0	0	k
sumCoef						1		

sumSize

	5	4	3	2	1	0	1	i
addendExpon				14	8	0		

$$3x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	1	i
addendCoef				3	-2	1		

addendSize

3

	5	4	3	2	1	0	0	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	0	j
adderCoef				8	-3	10		

adderSize

3

	5	4	3	2	1	0	1	k
sumExpon						0		

1

	5	4	3	2	1	0	1	k
sumCoef						1		

sumSize

	5	4	3	2	1	0	1	i
addendExpon				14	8	0		

$$3x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	1	i
addendCoef				3	-2	1		

$$\text{addendSize } \div 3$$

	5	4	3	2	1	0	0	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	0	j
adderCoef				8	-3	10		

$$\text{adderSize } \div 3$$

	5	4	3	2	1	0	1	k
sumExpon					6	0		

$$10x^6 + 1$$

	5	4	3	2	1	0	1	k
sumCoef					10	1		

$$\text{sumSize } \div$$

	5	4	3	2	1	0	1	i
addendExpon				14	8	0		

$$3x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	1	i
addendCoef				3	-2	1		

$$\text{addendSize } \div 3$$

	5	4	3	2	1	0	1	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	1	j
adderCoef				8	-3	10		

$$\text{adderSize } \div 3$$

	5	4	3	2	1	0	2	k
sumExpon					6	0		

$$10x^6 + 1$$

	5	4	3	2	1	0	2	k
sumCoef					10	1		

$$\text{sumSize } \div$$

	5	4	3	2	1	0	1	i
addendExpon				14	8	0		

$$3x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	1	i
addendCoef				3	-2	1		

$$\text{addendSize } \div 3$$

	5	4	3	2	1	0	1	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	1	j
adderCoef				8	-3	10		

$$\text{adderSize } \div 3$$

	5	4	3	2	1	0	2	k
sumExpon				8	6	0		

$$-2x^8 + 10x^6 + 1$$

	5	4	3	2	1	0	2	k
sumCoef				-2	10	1		

$$\text{sumSize } \div$$

	5	4	3	2	1	0	2	i
addendExpon				14	8	0		

$$3x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	2	i
addendCoef				3	-2	1		

$$\text{addendSize } \div 3$$

	5	4	3	2	1	0	1	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	1	j
adderCoef				8	-3	10		

$$\text{adderSize } \div 3$$

	5	4	3	2	1	0	3	k
sumExpon				8	6	0		

$$-2x^8 + 10x^6 + 1$$

	5	4	3	2	1	0	3	k
sumCoef				-2	10	1		

$$\text{sumSize } \div$$

	5	4	3	2	1	0	2	i
addendExpon				14	8	0		

$$3x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	2	i
addendCoef				3	-2	1		

$$\text{addendSize } \div 3$$

	5	4	3	2	1	0	1	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	1	j
adderCoef				8	-3	10		

$$\text{adderSize } \div 3$$

	5	4	3	2	1	0	3	k
sumExpon			10	8	6	0		

$$-3x^{10} - 2x^8 + 10x^6 + 1$$

	5	4	3	2	1	0	3	k
sumCoef			-3	-2	10	1		

$$\text{sumSize } \div$$

	5	4	3	2	1	0	2	i
addendExpon				14	8	0		

$$3x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	2	i
addendCoef				3	-2	1		

addendSize

3

	5	4	3	2	1	0	2	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	2	j
adderCoef				8	-3	10		

adderSize

3

	5	4	3	2	1	0	4	k
sumExpon			10	8	6	0		

$$-3x^{10} - 2x^8 + 10x^6 + 1$$

	5	4	3	2	1	0	4	k
sumCoef			-3	-2	10	1		

sumSize

	5	4	3	2	1	0	2	i
addendExpon				14	8	0		

$$3x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	2	i
addendCoef				3	-2	1		

addendSize

3

	5	4	3	2	1	0	2	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	2	j
adderCoef				8	-3	10		

adderSize

3

	5	4	3	2	1	0	4	k
sumExpon		14	10	8	6	0		

$$11x^{14} - 3x^{10} - 2x^8 + 10x^6 + 1$$

	5	4	3	2	1	0	4	k
sumCoef		11	-3	-2	10	1		

sumSize

	5	4	3	2	1	0	3	i
addendExpon				14	8	0		

$$3x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	3	i
addendCoef				3	-2	1		

addendSize

3

	5	4	3	2	1	0	3	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	3	j
adderCoef				8	-3	10		

adderSize

3

	5	4	3	2	1	0	5	k
sumExpon		14	10	8	6	0		

$$11x^{14} - 3x^{10} - 2x^8 + 10x^6 + 1$$

	5	4	3	2	1	0	5	k
sumCoef		11	-3	-2	10	1		

sumSize

	5	4	3	2	1	0	3 i
addendExpon				14	8	0	

$$3x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	3 i
addendCoef				3	-2	1	

addendSize

3

	5	4	3	2	1	0	3 j
adderExpon				14	10	6	

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	3 j
adderCoef				8	-3	10	

adderSize

3

	5	4	3	2	1	0	5 k
sumExpon		14	10	8	6	0	

$$11x^{14} - 3x^{10} - 2x^8 + 10x^6 + 1$$

	5	4	3	2	1	0	5 k
sumCoef		11	-3	-2	10	1	

sumSize

5

	5	4	3	2	1	0	
addendExpon				14	8	0	0 i

$$-8x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	
addendCoef				-8	-2	1	0 i

addendSize

3

	5	4	3	2	1	0	
adderExpon				14	10	6	0 j

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	
adderCoef				8	-3	10	0 j

adderSize

3

	5	4	3	2	1	0	
sumExpon							0 k

	5	4	3	2	1	0	
sumCoef							0 k

sumSize

	5	4	3	2	1	0	0	i
addendExpon				14	8	0		

$$-8x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	0	i
addendCoef				-8	-2	1		

addendSize 3

	5	4	3	2	1	0	0	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	0	j
adderCoef				8	-3	10		

adderSize 3

	5	4	3	2	1	0	0	k
sumExpon						0		

1

	5	4	3	2	1	0	0	k
sumCoef						1		

sumSize

	5	4	3	2	1	0	1	i
addendExpon				14	8	0		

$$-8x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	1	i
addendCoef				-8	-2	1		

addendSize

3

	5	4	3	2	1	0	0	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	0	j
adderCoef				8	-3	10		

adderSize

3

	5	4	3	2	1	0	1	k
sumExpon						0		

1

	5	4	3	2	1	0	1	k
sumCoef						1		

sumSize

	5	4	3	2	1	0	1	i
addendExpon				14	8	0		

$$-8x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	1	i
addendCoef				-8	-2	1		

$$\text{addendSize } \div 3$$

	5	4	3	2	1	0	0	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	0	j
adderCoef				8	-3	10		

$$\text{adderSize } \div 3$$

	5	4	3	2	1	0	1	k
sumExpon					6	0		

$$10x^6 + 1$$

	5	4	3	2	1	0	1	k
sumCoef					10	1		

$$\text{sumSize } \div$$

	5	4	3	2	1	0	1	i
addendExpon				14	8	0		

$$-8x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	1	i
addendCoef				-8	-2	1		

$$\text{addendSize } 3$$

	5	4	3	2	1	0	1	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	1	j
adderCoef				8	-3	10		

$$\text{adderSize } 3$$

	5	4	3	2	1	0	2	k
sumExpon					6	0		

$$10x^6 + 1$$

	5	4	3	2	1	0	2	k
sumCoef					10	1		

$$\text{sumSize } \boxed{}$$

	5	4	3	2	1	0	1	i
addendExpon				14	8	0		

$$-8x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	1	i
addendCoef				-8	-2	1		

addendSize

3

	5	4	3	2	1	0	1	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	1	j
adderCoef				8	-3	10		

adderSize

3

	5	4	3	2	1	0	2	k
sumExpon				8	6	0		

$$-2x^8 + 10x^6 + 1$$

	5	4	3	2	1	0	2	k
sumCoef				-2	10	1		

sumSize

	5	4	3	2	1	0	2	i
addendExpon				14	8	0		

$$-8x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	2	i
addendCoef				-8	-2	1		

addendSize

3

	5	4	3	2	1	0	1	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	1	j
adderCoef				8	-3	10		

adderSize

3

	5	4	3	2	1	0	3	k
sumExpon				8	6	0		

$$-2x^8 + 10x^6 + 1$$

	5	4	3	2	1	0	3	k
sumCoef				-2	10	1		

sumSize

	5	4	3	2	1	0	2	i
addendExpon				14	8	0		

$$-8x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	2	i
addendCoef				-8	-2	1		

$$\text{addendSize } \div 3$$

	5	4	3	2	1	0	1	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	1	j
adderCoef				8	-3	10		

$$\text{adderSize } \div 3$$

	5	4	3	2	1	0	3	k
sumExpon			10	8	6	0		

$$-3x^{10} - 2x^8 + 10x^6 + 1$$

	5	4	3	2	1	0	3	k
sumCoef			-3	-2	10	1		

$$\text{sumSize } \div$$

	5	4	3	2	1	0	2	i
addendExpon				14	8	0		

$$-8x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	2	i
addendCoef				-8	-2	1		

$$\text{addendSize } \div 3$$

	5	4	3	2	1	0	2	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	2	j
adderCoef				8	-3	10		

$$\text{adderSize } \div 3$$

	5	4	3	2	1	0	4	k
sumExpon			10	8	6	0		

$$-3x^{10} - 2x^8 + 10x^6 + 1$$

	5	4	3	2	1	0	4	k
sumCoef			-3	-2	10	1		

$$\text{sumSize } \div$$

	5	4	3	2	1	0
addendExpon				14	8	0

2

 i

$$-8x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0
addendCoef				-8	-2	1

2

 i

addendSize

3

	5	4	3	2	1	0
adderExpon				14	10	6

2

 j

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0
adderCoef				8	-3	10

2

 j

adderSize

3

	5	4	3	2	1	0
sumExpon			10	8	6	0

4

 k

$$-3x^{10} - 2x^8 + 10x^6 + 1$$

	5	4	3	2	1	0
sumCoef		0	-3	-2	10	1

4

 k

sumSize

	5	4	3	2	1	0
addendExpon				14	8	0

3

 i

$$-8x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0
addendCoef				-8	-2	1

3

 i

 addendSize

3

	5	4	3	2	1	0
adderExpon				14	10	6

3

 j

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0
adderCoef				8	-3	10

3

 j

 adderSize

3

	5	4	3	2	1	0
sumExpon			10	8	6	0

4

 k

$$-3x^{10} - 2x^8 + 10x^6 + 1$$

	5	4	3	2	1	0
sumCoef		0	-3	-2	10	1

4

 k

 sumSize

	5	4	3	2	1	0	3	i
addendExpon				14	8	0		

$$-8x^{14} - 2x^8 + 1$$

	5	4	3	2	1	0	3	i
addendCoef				-8	-2	1		

$$\text{addendSize } 3$$

	5	4	3	2	1	0	3	j
adderExpon				14	10	6		

$$8x^{14} - 3x^{10} + 10x^6$$

	5	4	3	2	1	0	3	j
adderCoef				8	-3	10		

$$\text{adderSize } 3$$

	5	4	3	2	1	0	4	k
sumExpon			10	8	6	0		

$$-3x^{10} - 2x^8 + 10x^6 + 1$$

	5	4	3	2	1	0	4	k
sumCoef		0	-3	-2	10	1		

$$\text{sumSize } 4$$

Multiplication

```
product = 0;
if( multiplicand != 0 && multiplier != 0 )
    for( int i = 0; i < multiplierSize; i++ )
    {
        buffer = multiplier[ i ] * multiplicand
        product += buffer;
    }
```

	7	6	5	4	3	2	1	0
multiplicandExpon						14	8	2

multiplicandCoef						3	-2	1
------------------	--	--	--	--	--	---	----	---

$$3x^{14} - 2x^8 + x^2$$

multiplicandSize 3

	7	6	5	4	3	2	1	0
multiplierExpon						16	10	4

multiplierCoef						2	2	2
----------------	--	--	--	--	--	---	---	---

$$2x^{16} + 2x^{10} + 2x^4$$

multiplierSize 3

	7	6	5	4	3	2	1	0
bufferExpon								

bufferCoef								
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bufferSize

	7	6	5	4	3	2	1	0
productExpon								0

productCoef								0
-------------	--	--	--	--	--	--	--	---

productSize 1

	7	6	5	4	3	2	1	0
multiplicandExpon						14	8	2

multiplicandCoef						3	-2	1
------------------	--	--	--	--	--	---	----	---

$$3x^{14} - 2x^8 + x^2$$

multiplicandSize 3

	7	6	5	4	3	2	1	0
multiplierExpon						16	10	4

multiplierCoef						2	2	2
----------------	--	--	--	--	--	---	---	---

$$2x^{16} + 2x^{10} + 2x^4$$

multiplierSize 3

	7	6	5	4	3	2	1	0
bufferExpon						18	12	6

bufferCoef						6	-4	2
------------	--	--	--	--	--	---	----	---

$$6x^{18} - 4x^{12} + 2x^6$$

bufferSize 3

	7	6	5	4	3	2	1	0
productExpon								0

productCoef								0
-------------	--	--	--	--	--	--	--	---

productSize 1

	7	6	5	4	3	2	1	0
multiplicandExpon						14	8	2

multiplicandCoef						3	-2	1
------------------	--	--	--	--	--	---	----	---

$$3x^{14} - 2x^8 + x^2$$

multiplicandSize 3

	7	6	5	4	3	2	1	0
multiplierExpon						16	10	4

multiplierCoef						2	2	2
----------------	--	--	--	--	--	---	---	---

$$2x^{16} + 2x^{10} + 2x^4$$

multiplierSize 3

	7	6	5	4	3	2	1	0
bufferExpon						18	12	6

bufferCoef						6	-4	2
------------	--	--	--	--	--	---	----	---

$$6x^{18} - 4x^{12} + 2x^6$$

bufferSize 3

	7	6	5	4	3	2	1	0
productExpon						18	12	6

productCoef						6	-4	2
-------------	--	--	--	--	--	---	----	---

$$6x^{18} - 4x^{12} + 2x^6$$

productSize 3

	7	6	5	4	3	2	1	0
multiplicandExpon						14	8	2

multiplicandCoef						3	-2	1
------------------	--	--	--	--	--	---	----	---

$$3x^{14} - 2x^8 + x^2$$

multiplicandSize 3

	7	6	5	4	3	2	1	0
multiplierExpon						16	10	4

multiplierCoef						2	2	2
----------------	--	--	--	--	--	---	---	---

$$2x^{16} + 2x^{10} + 2x^4$$

multiplierSize 3

	7	6	5	4	3	2	1	0
bufferExpon						24	18	12

bufferCoef						6	-4	2
------------	--	--	--	--	--	---	----	---

$$6x^{24} - 4x^{18} + 2x^{12}$$

bufferSize 3

	7	6	5	4	3	2	1	0
productExpon						18	12	6

productCoef						6	-4	2
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$$6x^{18} - 4x^{12} + 2x^6$$

productSize 3

	7	6	5	4	3	2	1	0
multiplicandExpon						14	8	2
multiplicandCoef						3	-2	1

$$3x^{14} - 2x^8 + x^2$$

multiplicandSize 3

	7	6	5	4	3	2	1	0
multiplierExpon						16	10	4
multiplierCoef						2	2	2

$$2x^{16} + 2x^{10} + 2x^4$$

multiplierSize 3

	7	6	5	4	3	2	1	0
bufferExpon						24	18	12
bufferCoef						6	-4	2

$$6x^{24} - 4x^{18} + 2x^{12}$$

bufferSize 3

	7	6	5	4	3	2	1	0
productExpon					24	18	12	6
productCoef					6	2	-2	2

$$6x^{24} + 2x^{18} - 2x^{12} + 2x^6$$

productSize 4

	7	6	5	4	3	2	1	0
multiplicandExpon						14	8	2

multiplicandCoef						3	-2	1
------------------	--	--	--	--	--	---	----	---

$$3x^{14} - 2x^8 + x^2$$

multiplicandSize 3

	7	6	5	4	3	2	1	0
multiplierExpon						16	10	4

multiplierCoef						2	2	2
----------------	--	--	--	--	--	---	---	---

$$2x^{16} + 2x^{10} + 2x^4$$

multiplierSize 3

	7	6	5	4	3	2	1	0
bufferExpon						30	24	18

bufferCoef						6	-4	2
------------	--	--	--	--	--	---	----	---

$$6x^{30} - 4x^{24} + 2x^{18}$$

bufferSize 3

	7	6	5	4	3	2	1	0
productExpon					24	18	12	6

productCoef					6	2	-2	2
-------------	--	--	--	--	---	---	----	---

$$6x^{24} + 2x^{18} - 2x^{12} + 2x^6$$

productSize 4

	7	6	5	4	3	2	1	0
multiplicandExpon						14	8	2
multiplicandCoef						3	-2	1

$$3x^{14} - 2x^8 + x^2$$

multiplicandSize 3

	7	6	5	4	3	2	1	0
multiplierExpon						16	10	4
multiplierCoef						2	2	2

$$2x^{16} + 2x^{10} + 2x^4$$

multiplierSize 3

	7	6	5	4	3	2	1	0
bufferExpon						30	24	18
bufferCoef						6	-4	2

$$6x^{30} - 4x^{24} + 2x^{18}$$

bufferSize 3

	7	6	5	4	3	2	1	0
productExpon				30	24	18	12	6
productCoef				6	2	4	-2	2

$$6x^{30} + 2x^{24} + 4x^{18} - 2x^{12} + 2x^6$$

productSize 5

Division

```

remainder = dividend;
int i;
for( i = quotientSize - 1; i >= 0; i-- )
{
    quotient[ i ] = remainder[ remainderSize - 1 ] /
                    divisor[ divisorSize - 1 ];

    buffer = divisor * quotient[ i ];
    if( remainder == buffer )
    {
        remainder = 0;
        break;
    }

    remainder -= buffer

    if( the degree of remainder < the degree of divisor )
        break;
}

right shift quotient over i positions (terms);

```

remainderSize 3

	3	2	1	0
remainderExpon		3	2	0
remainderCoef		1	-2	-4

divisorSize 2

	3	2	1	0
divisorExpon			1	0
divisorCoef			1	-3

bufferSize

	3	2	1	0
bufferExpon				
bufferCoef				

```
remainder = dividend;  
int i;
```

$$x - 3 \overline{) x^3 - 2x^2 + 0x - 4}$$

quotientSize 20

	19	18	17	16
quotientExpon				
quotientCoef				

remainderSize 3

	3	2	1	0
remainderExpon		3	2	0

remainderCoef		1	-2	-4
---------------	--	---	--	--

divisorSize 2

	3	2	1	0
divisorExpon			1	0

divisorCoef			1	-3
-------------	--	--	---	--

bufferSize

	3	2	1	0
bufferExpon				

bufferCoef				
------------	--	--	--	--

$$x - 3 \overline{) x^3 - 2x^2 + 0x - 4}$$

quotientSize 20

	i			
	19	18	17	16
quotientExpon				
quotientCoef				

```
for( i = quotientSize - 1; i >= 0; i-- ) {  
    quotient[ i ] = remainder[ remainderSize - 1 ] /  
                    divisor[ divisorSize - 1 ];  
    buffer = divisor * quotient[ i ];  
    if( remainder == buffer ) {  
        remainder = 0; break;  
    }  
    remainder -= buffer;  
}
```

remainderSize 3

	3	2	1	0
remainderExpon		3	2	0
remainderCoef		1	-2	-4

$$\begin{array}{r}
 x^2 \\
 x - 3 \overline{) x^3 - 2x^2 + 0x - 4}
 \end{array}$$

divisorSize 2

	3	2	1	0
divisorExpon			1	0
divisorCoef			1	-3

bufferSize

	3	2	1	0
bufferExpon				
bufferCoef				

quotientSize 20

	i			
	19	18	17	16
quotientExpon	2			
quotientCoef	1			

```

for( i = quotientSize - 1; i >= 0; i-- ) {
    quotient[ i ] = remainder[ remainderSize - 1 ] /
                    divisor[ divisorSize - 1 ];
    buffer = divisor * quotient[ i ];
    if( remainder == buffer ) {
        remainder = 0; break;
    }
    remainder -= buffer;
}

```


remainderSize 3

	3	2	1	0
remainderExpon		3	2	0
remainderCoef		1	-2	-4

$$\begin{array}{r}
 x^2 \\
 x - 3 \overline{) \quad x^3 - 2x^2 + 0x - 4} \\
 \underline{x^3 - 3x^2} \\
 3x^2 + 0x - 4
 \end{array}$$

divisorSize 2

	3	2	1	0
divisorExpon			1	0
divisorCoef			1	-3

bufferSize 2

	3	2	1	0
bufferExpon			3	2
bufferCoef			1	-3

quotientSize 20

	i			
	19	18	17	16
quotientExpon	2			
quotientCoef	1			

```

for( i = quotientSize - 1; i >= 0; i-- ) {
    quotient[ i ] = remainder[ remainderSize - 1 ] /
                    divisor[ divisorSize - 1 ];
    buffer = divisor * quotient[ i ];
    if( remainder == buffer ) {
        remainder = 0; break;
    }
    remainder -= buffer;
}

```

remainderSize 2

	3	2	1	0
remainderExpon			2	0
remainderCoef			1	-4

$$\begin{array}{r}
 x^2 \\
 x - 3 \overline{) \begin{array}{l} x^3 - 2x^2 + 0x - 4 \\ x^3 - 3x^2 \\ \hline x^2 + 0x - 4 \end{array}}
 \end{array}$$

divisorSize 2

	3	2	1	0
divisorExpon			1	0
divisorCoef			1	-3

bufferSize 2

	3	2	1	0
bufferExpon			3	2
bufferCoef			1	-3

quotientSize 20

	i			
	19	18	17	16
quotientExpon	2			
quotientCoef	1			

```

for( i = quotientSize - 1; i >= 0; i-- ) {
    quotient[ i ] = remainder[ remainderSize - 1 ] /
                    divisor[ divisorSize - 1 ];
    buffer = divisor * quotient[ i ];
    if( remainder == buffer ) {
        remainder = 0; break;
    }
    remainder -= buffer;
}

```

remainderSize 2

	3	2	1	0
remainderExpon			2	0
remainderCoef			1	-4

$$\begin{array}{r}
 x^2 + x \\
 x - 3 \overline{) \begin{array}{l} x^3 - 2x^2 + 0x - 4 \\ x^3 - 3x^2 \end{array}} \\
 \hline
 x^2 + 0x - 4
 \end{array}$$

divisorSize 2

	3	2	1	0
divisorExpon			1	0
divisorCoef			1	-3

bufferSize 2

	3	2	1	0
bufferExpon			3	2
bufferCoef			1	-3

quotientSize 20

		i		
	19	18	17	16
quotientExpon	2	1		
quotientCoef	1	1		

```

for( i = quotientSize - 1; i >= 0; i-- ) {
    quotient[ i ] = remainder[ remainderSize - 1 ] /
                    divisor[ divisorSize - 1 ];
    buffer = divisor * quotient[ i ];
    if( remainder == buffer ) {
        remainder = 0; break;
    }
    remainder -= buffer;
}

```

remainderSize 2

	3	2	1	0
remainderExpon			2	0
remainderCoef			1	-4

divisorSize 2

	3	2	1	0
divisorExpon			1	0
divisorCoef			1	-3

bufferSize 2

	3	2	1	0
bufferExpon			2	1
bufferCoef			1	-3

$$\begin{array}{r}
 x^2 + x \\
 x - 3 \overline{) \begin{array}{l} x^3 - 2x^2 + 0x - 4 \\ x^3 - 3x^2 \end{array}} \\
 \hline
 x^2 + 0x - 4 \\
 x^2 - 3x
 \end{array}$$

quotientSize 20

	19	18	17	16
quotientExpon	2	1		
quotientCoef	1	1		

```

for( i = quotientSize - 1; i >= 0; i-- ) {
    quotient[ i ] = remainder[ remainderSize - 1 ] /
                    divisor[ divisorSize - 1 ];
    buffer = divisor * quotient[ i ];
    if( remainder == buffer ) {
        remainder = 0; break;
    }
    remainder -= buffer;

```

remainderSize 2

3	2	1	0
		1	0

remainderCoef

		3	-4
		3	-4

divisorSize 2

3	2	1	0
		1	0

divisorExpon

		1	-3
		1	-3

divisorCoef

bufferSize 2

3	2	1	0
		2	1

bufferExpon

		1	-3
		1	-3

bufferCoef

$$\begin{array}{r}
 x^2 + x \\
 x - 3 \overline{) \begin{array}{l} x^3 - 2x^2 + 0x - 4 \\ x^3 - 3x^2 \end{array}} \\
 \hline
 \begin{array}{l} x^2 + 0x - 4 \\ x^2 - 3x \end{array} \\
 \hline
 3x - 4
 \end{array}$$

quotientSize 20

		i		
19	18	17	16	

quotientExpon

2	1		
---	---	--	--

quotientCoef

1	1		
---	---	--	--

```

for( i = quotientSize - 1; i >= 0; i-- ) {
    quotient[ i ] = remainder[ remainderSize - 1 ] /
                    divisor[ divisorSize - 1 ];
    buffer = divisor * quotient[ i ];
    if( remainder == buffer ) {
        remainder = 0; break;
    }
    remainder -= buffer;

```

remainderSize 2

3	2	1	0
		1	0

remainderCoef

		3	-4
		3	-4

divisorSize 2

3	2	1	0
		1	0

divisorExpon

		1	-3
		1	-3

divisorCoef

bufferSize 2

3	2	1	0
		2	1

bufferExpon

		1	-3
		1	-3

bufferCoef

$$\begin{array}{r}
 x^2 + x + 3 \\
 x - 3 \overline{) \begin{array}{l} x^3 - 2x^2 + 0x - 4 \\ x^3 - 3x^2 \end{array}} \\
 \hline
 x^2 + 0x - 4 \\
 x^2 - 3x \\
 \hline
 3x - 4
 \end{array}$$

quotientSize 20

19	18	17	16
2	1	0	

quotientExpon

1	1	3	
1	1	3	

quotientCoef

```

for( i = quotientSize - 1; i >= 0; i-- ) {
    quotient[ i ] = remainder[ remainderSize - 1 ] /
                    divisor[ divisorSize - 1 ];
    buffer = divisor * quotient[ i ];
    if( remainder == buffer ) {
        remainder = 0; break;
    }
    remainder -= buffer;

```

remainderSize 2

3	2	1	0
		1	0

remainderCoef

		3	-4
		3	-4

divisorSize 2

3	2	1	0
		1	0

divisorExpon

		1	-3
		1	-3

divisorCoef

bufferSize 2

3	2	1	0
		1	0

bufferExpon

		3	-9
		3	-9

bufferCoef

$$\begin{array}{r}
 x^2 + x + 3 \\
 x - 3 \overline{) \begin{array}{l} x^3 - 2x^2 + 0x - 4 \\ x^3 - 3x^2 \end{array}} \\
 \hline
 \begin{array}{l} x^2 + 0x - 4 \\ x^2 - 3x \end{array} \\
 \hline
 \begin{array}{l} 3x - 4 \\ 3x - 9 \end{array}
 \end{array}$$

quotientSize 20

19	18	17	16
2	1	0	

quotientExpon

1	1	3	
1	1	3	

quotientCoef

```

for( i = quotientSize - 1; i >= 0; i-- ) {
    quotient[ i ] = remainder[ remainderSize - 1 ] /
                    divisor[ divisorSize - 1 ];
    buffer = divisor * quotient[ i ];
    if( remainder == buffer ) {
        remainder = 0; break;
    }
    remainder -= buffer;

```

remainderSize 1

	3	2	1	0
remainderExpon				0
remainderCoef				5

divisorSize 2

	3	2	1	0
divisorExpon			1	0
divisorCoef			1	-3

bufferSize 2

	3	2	1	0
bufferExpon			1	0
bufferCoef			3	-9

$$\begin{array}{r}
 x^2 + x + 3 \\
 x - 3 \overline{) \begin{array}{l} x^3 - 2x^2 + 0x - 4 \\ x^3 - 3x^2 \\ \hline x^2 + 0x - 4 \\ x^2 - 3x \\ \hline 3x - 4 \\ 3x - 9 \\ \hline 5 \end{array} }
 \end{array}$$

quotientSize 20

	19	18	17	16
quotientExpon	2	1	0	
quotientCoef	1	1	3	

```

for( i = quotientSize - 1; i >= 0; i-- ) {
    quotient[ i ] = remainder[ remainderSize - 1 ] /
                    divisor[ divisorSize - 1 ];
    buffer = divisor * quotient[ i ];
    if( remainder == buffer ) {
        remainder = 0; break;
    }
    remainder -= buffer;
}

```


remainderSize 1

	3	2	1	0
remainderExpon				0
remainderCoef				5

divisorSize 2

	3	2	1	0
divisorExpon			1	0
divisorCoef			1	-3

bufferSize 2

	3	2	1	0
bufferExpon			1	0
bufferCoef			3	-9

$$\begin{array}{r}
 x^2 + x + 3 \\
 x - 3 \overline{) \begin{array}{l} x^3 - 2x^2 + 0x - 4 \\ x^3 - 3x^2 \\ \hline x^2 + 0x - 4 \\ x^2 - 3x \\ \hline 3x - 4 \\ 3x - 9 \\ \hline 5 \end{array}}
 \end{array}$$

quotientSize 20

	19	18	i	17	16
quotientExpon	2	1	0		
quotientCoef	1	1	3		

```

    if( the degree of remainder < the degree of divisor )
        break;
}

```

right shift quotient over i positions (terms);

remainderSize 1

	3	2	1	0
remainderExpon				0
remainderCoef				5

divisorSize 2

	3	2	1	0
divisorExpon			1	0
divisorCoef			1	-3

bufferSize 2

	3	2	1	0
bufferExpon			1	0
bufferCoef			3	-9

$$\begin{array}{r}
 x^2 + x + 3 \\
 x - 3 \overline{) \begin{array}{l} x^3 - 2x^2 + 0x - 4 \\ x^3 - 3x^2 \\ \hline x^2 + 0x - 4 \\ x^2 - 3x \\ \hline 3x - 4 \\ 3x - 9 \\ \hline 5 \end{array} }
 \end{array}$$

quotientSize 3

	2	1	i
quotientExpon	2	1	0
quotientCoef	1	1	3

```

    if( the degree of remainder < the degree of divisor )
        break;
}

right shift quotient over i positions (terms);

```


remainderSize 3

	3	2	1	0
remainderExpon		3	2	1
remainderCoef		1	-2	-3

divisorSize 2

	3	2	1	0
divisorExpon			1	0
divisorCoef			1	-3

bufferSize

	3	2	1	0
bufferExpon				
bufferCoef				

quotientSize 20

	19	18	17	16
quotientExpon				
quotientCoef				

$$x - 3 \overline{) x^3 - 2x^2 - 3x}$$

```
remainder = dividend;  
int i;
```

remainderSize 3

	3	2	1	0
remainderExpon		3	2	1

remainderCoef		1	-2	-3
---------------	--	---	--	--

divisorSize 2

	3	2	1	0
divisorExpon			1	0

divisorCoef			1	-3
-------------	--	--	---	--

bufferSize

	3	2	1	0
bufferExpon				

bufferCoef				
------------	--	--	--	--

$$x - 3 \overline{) x^3 - 2x^2 - 3x}$$

quotientSize 20

	i			
	19	18	17	16
quotientExpon				
quotientCoef				

```
for( i = quotientSize - 1; i >= 0; i-- ) {  
    quotient[ i ] = remainder[ remainderSize - 1 ] /  
                    divisor[ divisorSize - 1 ];  
    buffer = divisor * quotient[ i ];  
    if( remainder == buffer ) {  
        remainder = 0; break;  
    }  
    remainder -= buffer;  
}
```

remainderSize 3

	3	2	1	0
remainderExpon		3	2	1

remainderCoef		1	-2	-3
---------------	--	---	--	--

divisorSize 2

	3	2	1	0
divisorExpon			1	0

divisorCoef			1	-3
-------------	--	--	---	--

bufferSize

	3	2	1	0
bufferExpon				

bufferCoef				
------------	--	--	--	--

$$\begin{array}{r} x^2 \\ x - 3 \overline{) x^3 - 2x^2 - 3x} \end{array}$$

quotientSize	20	i			
		19	18	17	16
quotientExpon		2			
quotientCoef		1			

```
for( i = quotientSize - 1; i >= 0; i-- ) {  
    quotient[ i ] = remainder[ remainderSize - 1 ] /  
                    divisor[ divisorSize - 1 ];  
    buffer = divisor * quotient[ i ];  
    if( remainder == buffer ) {  
        remainder = 0; break;  
    }  
    remainder -= buffer;  
}
```

remainderSize 3

	3	2	1	0
remainderExpon		3	2	1
remainderCoef		1	-2	-3

$$\begin{array}{r}
 x^2 \\
 x - 3 \overline{) \quad x^3 - 2x^2 - 3x} \\
 \underline{x^3 - 3x^2} \\
 3x^2 - 3x
 \end{array}$$

divisorSize 2

	3	2	1	0
divisorExpon			1	0
divisorCoef			1	-3

bufferSize 2

	3	2	1	0
bufferExpon			3	2
bufferCoef			1	-3

quotientSize 20

	i			
	19	18	17	16
quotientExpon	2			
quotientCoef	1			

```

for( i = quotientSize - 1; i >= 0; i-- ) {
    quotient[ i ] = remainder[ remainderSize - 1 ] /
                    divisor[ divisorSize - 1 ];
    buffer = divisor * quotient[ i ];
    if( remainder == buffer ) {
        remainder = 0; break;
    }
    remainder -= buffer;
}

```

remainderSize 2

	3	2	1	0
remainderExpon			2	1
remainderCoef			1	-3

$$\begin{array}{r}
 x^2 \\
 x - 3 \overline{) \begin{array}{l} x^3 - 2x^2 - 3x \\ x^3 - 3x^2 \\ \hline x^2 - 3x \end{array}}
 \end{array}$$

divisorSize 2

	3	2	1	0
divisorExpon			1	0
divisorCoef			1	-3

bufferSize 2

	3	2	1	0
bufferExpon			3	2
bufferCoef			1	-3

quotientSize 20

	i			
	19	18	17	16
quotientExpon	2			
quotientCoef	1			

```

for( i = quotientSize - 1; i >= 0; i-- ) {
    quotient[ i ] = remainder[ remainderSize - 1 ] /
                    divisor[ divisorSize - 1 ];
    buffer = divisor * quotient[ i ];
    if( remainder == buffer ) {
        remainder = 0; break;
    }
    remainder -= buffer;
}

```


remainderSize 2

	3	2	1	0
remainderExpon			2	1
remainderCoef			1	-3

$$\begin{array}{r}
 x^2 + x \\
 x - 3 \overline{) \begin{array}{l} x^3 - 2x^2 - 3x \\ x^3 - 3x^2 \\ \hline x^2 - 3x \end{array}}
 \end{array}$$

divisorSize 2

	3	2	1	0
divisorExpon			1	0
divisorCoef			1	-3

bufferSize 2

	3	2	1	0
bufferExpon			3	2
bufferCoef			1	-3

quotientSize 20

	19	18	17	16
quotientExpon	2	1		
quotientCoef	1	1		

```

for( i = quotientSize - 1; i >= 0; i-- ) {
    quotient[ i ] = remainder[ remainderSize - 1 ] /
                    divisor[ divisorSize - 1 ];
    buffer = divisor * quotient[ i ];
    if( remainder == buffer ) {
        remainder = 0; break;
    }
    remainder -= buffer;
}

```

remainderSize 2

3	2	1	0
		2	1

remainderCoef

		1	-3
--	--	---	----

divisorSize 2

3	2	1	0
		1	0

divisorCoef

		1	-3
--	--	---	----

bufferSize 2

3	2	1	0
		2	1

bufferCoef

		1	-3
--	--	---	----

$$\begin{array}{r}
 x^2 + x \\
 x - 3 \overline{) \begin{array}{l} x^3 - 2x^2 - 3x \\ x^3 - 3x^2 \\ \hline x^2 - 3x \\ x^2 - 3x \end{array} }
 \end{array}$$

quotientSize 20

19	18	17	16
2	1		

quotientCoef

1	1		
---	---	--	--

```

for( i = quotientSize - 1; i >= 0; i-- ) {
    quotient[ i ] = remainder[ remainderSize - 1 ] /
                    divisor[ divisorSize - 1 ];
    buffer = divisor * quotient[ i ];
    if( remainder == buffer ) {
        remainder = 0; break;
    }
    remainder -= buffer;

```

remainderSize 1

	3	2	1	0
remainderExpon				0

remainderCoef				0
---------------	-------------	-------------	-------------	--------------

divisorSize 2

	3	2	1	0
divisorExpon			1	0

divisorCoef			1	-3
-------------	-------------	-------------	--------------	---------------

bufferSize 2

	3	2	1	0
bufferExpon			2	1

bufferCoef			1	-3
------------	-------------	-------------	--------------	---------------

$$\begin{array}{r} x^2 + x \\ x - 3 \overline{) x^3 - 2x^2 - 3x} \\ \underline{x^3 - 3x^2} \\ x^2 - 3x \\ \underline{x^2 - 3x} \\ 0 \end{array}$$

quotientSize	20			
		i		
	19	18	17	16
quotientExpon	2	1		
quotientCoef	1	1		

```
for( i = quotientSize - 1; i >= 0; i-- ) {  
    quotient[ i ] = remainder[ remainderSize - 1 ] /  
                    divisor[ divisorSize - 1 ];  
    buffer = divisor * quotient[ i ];  
    if( remainder == buffer ) {  
        remainder = 0; break;  
    }  
    remainder -= buffer;  
}
```

remainderSize 1

	3	2	1	0
remainderExpon				0
remainderCoef				0

divisorSize 2

	3	2	1	0
divisorExpon			1	0
divisorCoef			1	-3

bufferSize 2

	3	2	1	0
bufferExpon			2	1
bufferCoef			1	-3

$$\begin{array}{r}
 x^2 + x \\
 \hline
 x - 3 \) \ x^3 - 2x^2 - 3x \\
 \underline{x^3 - 3x^2} \\
 x^2 - 3x \\
 \underline{x^2 - 3x} \\
 0
 \end{array}$$

quotientSize 20

		i		
	19	18	17	16
quotientExpon	2	1		
quotientCoef	1	1		

```

    if( the degree of remainder < the degree of divisor )
        break;
}

```

right shift quotient over i positions (terms);

remainderSize 1

	3	2	1	0
remainderExpon				0
remainderCoef				0

divisorSize 2

	3	2	1	0
divisorExpon			1	0
divisorCoef			1	-3

bufferSize 2

	3	2	1	0
bufferExpon			2	1
bufferCoef			1	-3

$$\begin{array}{r}
 x^2 + x \\
 \hline
 x - 3 \) \ x^3 - 2x^2 - 3x \\
 \underline{x^3 - 3x^2} \\
 x^2 - 3x \\
 \underline{x^2 - 3x} \\
 0
 \end{array}$$

quotientSize 2

	1	0
quotientExpon	2	1
quotientCoef	1	1

```

    if( the degree of remainder < the degree of divisor )
        break;
}

```

right shift quotient over i positions (terms);